

Transfer Learning vs. Random Initialization

“How much of transfer learning’s advantage is the pretrained features — and how much is just fewer knobs to turn?”

☒ Pretrained-Frozen

☐ Random-Frozen

☐ Random-Full

PRESENTER

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DATASET

Intel Image Classification • 6 classes

BACKBONE

ResNet-18

One comparison, two hidden variables

- The usual move: take an **ImageNet-pretrained** network, freeze the early layers, fine-tune the rest.
- It beats training from scratch — but **why?**
- Freezing changes **two** things at once: the **quality** of the frozen features and the **number** of trainable parameters.
- A naive pretrained-vs-scratch test **confounds** them — you can't tell which one earned the win.

FACTOR A

Feature quality

Pretrained weights encode edges, textures, shapes learned from a million images.

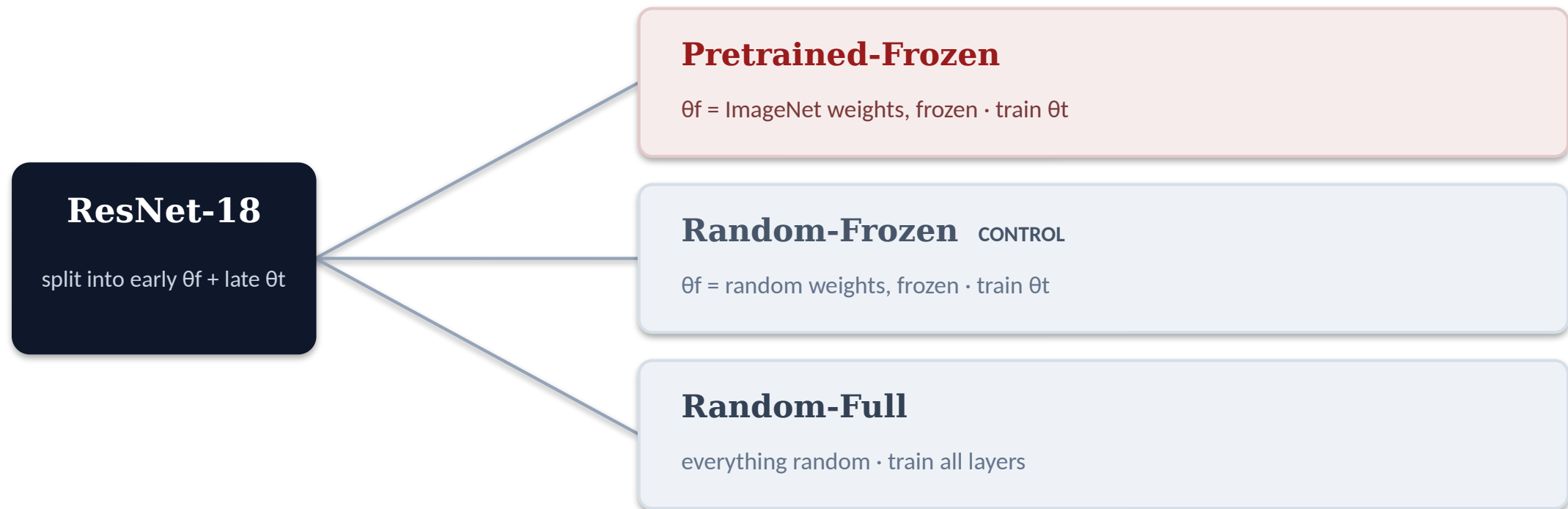
FACTOR B

Trainable capacity

Freezing early layers also shrinks how many parameters the optimizer can move.

Goal: a design that isolates Factor A from Factor B.

Three conditions, one controlled variable



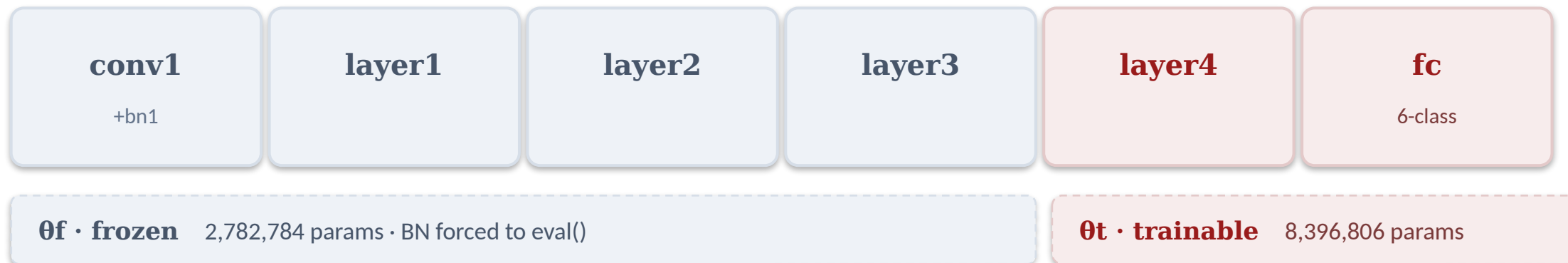
PT-F vs R-F → isolates **feature quality** (capacity matched)

R-F vs R-FL → isolates **trainable capacity** (both random)

METHOD

Where the network is cut

Freeze the generic early layers; train only the task-specific tail. PT-F and R-F share the **exact same** trainable count — that's what makes them a fair fight.



The subtle bug I avoided: if frozen BatchNorm layers keep updating their running stats, the representation drifts and training collapses. In both frozen conditions BN stays in eval mode. In Random-Full nothing is frozen, so BN updates normally — different by design, not by accident.

Same data, same recipe, three seeds

DATA

Intel Scenes

- 6 classes: buildings, forest, glacier, mountain, sea, street
- **12,631** train / **1,403** val / **3,000** test (held out)
- Resize 224, center-crop, ImageNet normalize

TRAINING

Identical for all

- SGD · lr **1e-3** · momentum **0.9** · wd **1e-4**
- Batch 32 · cross-entropy on θ_t
- Early stop on val loss, patience **5**

RIGOR

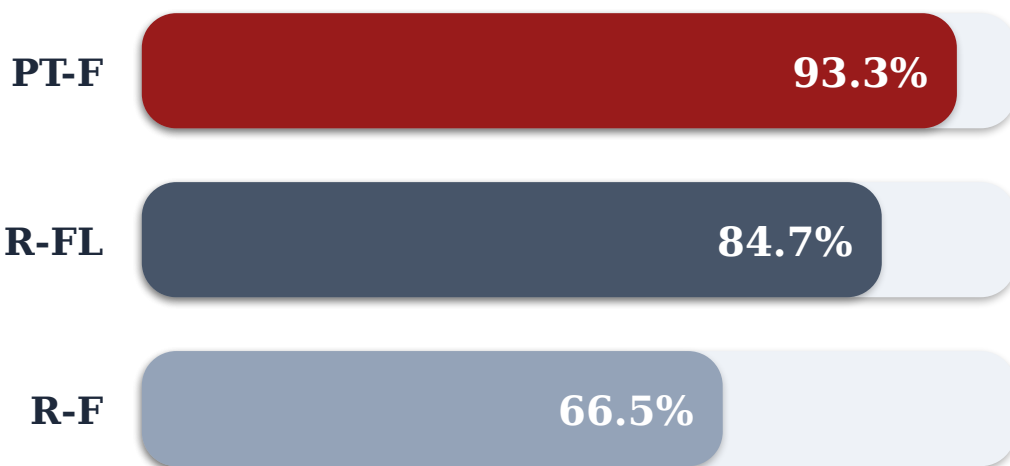
Reproducible

- 3 seeds: **42, 100, 2026**
- Test split touched **only once**, after model selection
- Report mean $\pm$ std, not single runs

RESULTS

Pretrained wins — on accuracy and speed

TEST ACCURACY • mean of 3 seeds



CONVERGENCE & COST

Pretrained-Frozen 8.0 epochs • ~103s

Random-Frozen 8.7 epochs • ~113s

Random-Full 15.0 epochs • ~426s

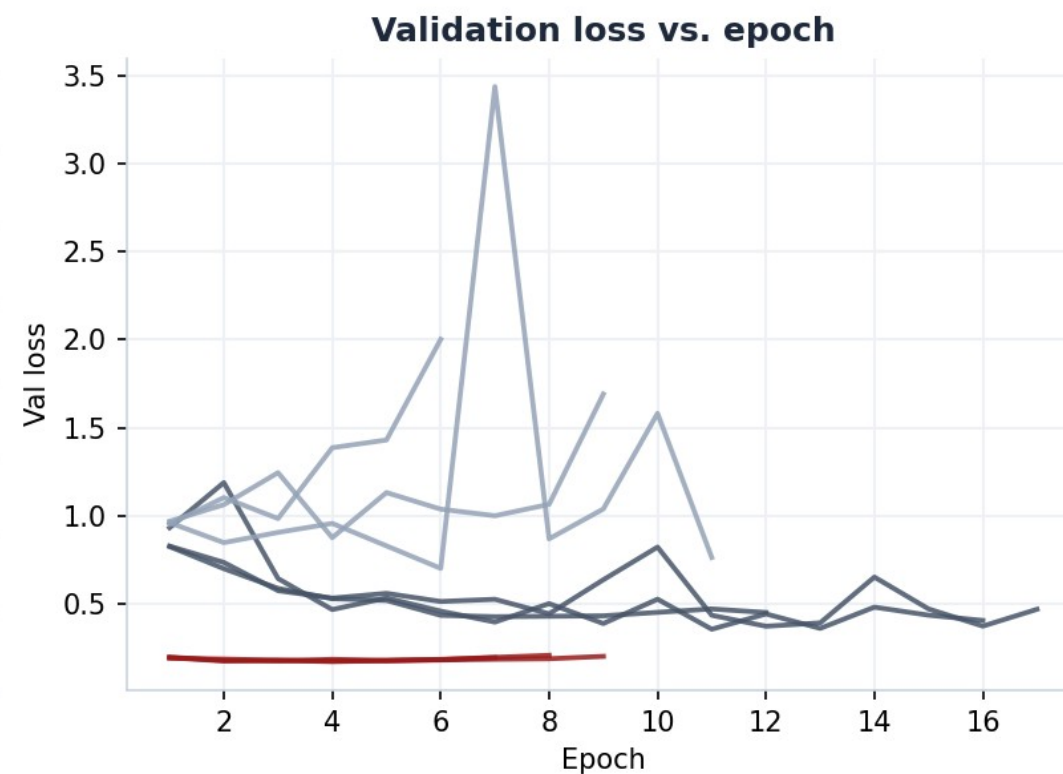
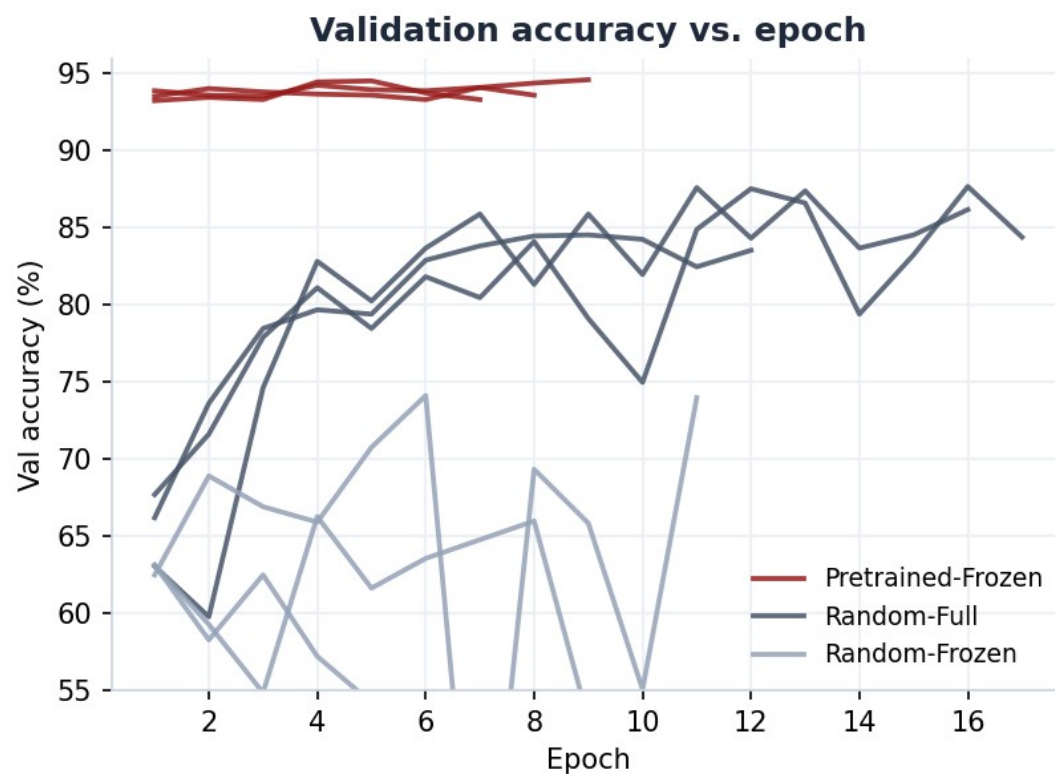
Training from scratch pays ~4× the wall-clock.

Capacity-matched gap: +26.7 points. PT-F vs R-F — identical trainable parameters and recipe, the only difference is pretrained vs random frozen features. That gap is pretraining, cleanly isolated. Unfreezing everything (R-FL) recovers accuracy but never reaches the pretrained ceiling.

RESULTS

Pretrained converges instantly; scratch crawls

Pretrained-frozen starts near its ceiling and flatlines in ~4 epochs; random-full climbs for 12–17 epochs; random-frozen is noisy and low.

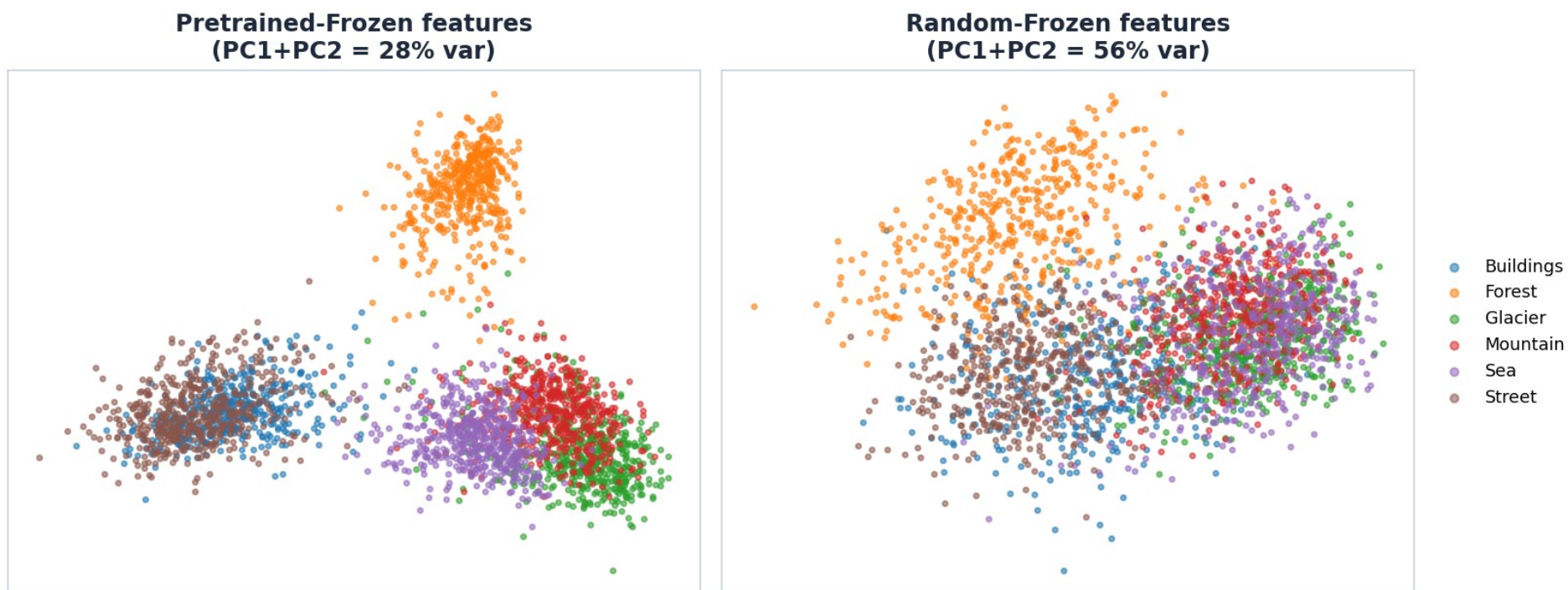


WHY IT WORKS

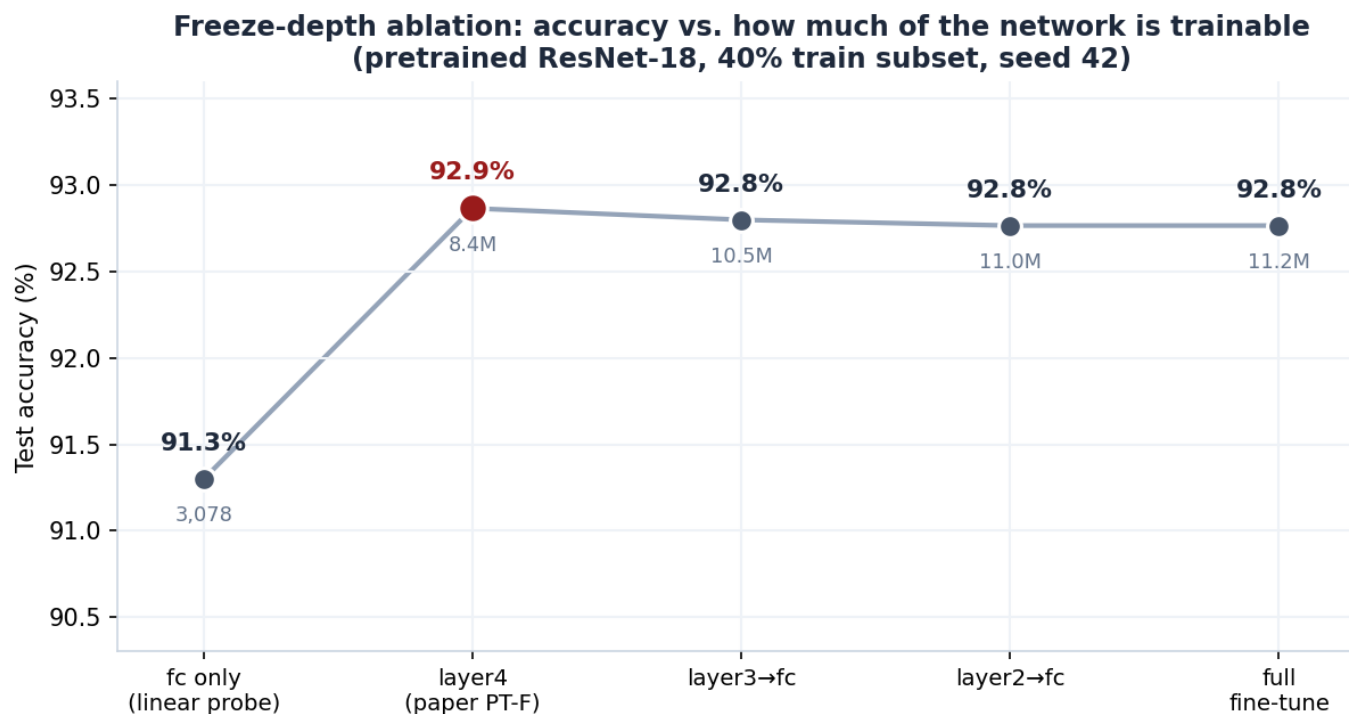
Pretrained features are already separable

PCA of the 512-d penultimate features: pretrained-frozen splits the classes (overlaps match the confusion matrix); random-frozen stays entangled.

PCA of penultimate (512-d) features on the test split · seed 42



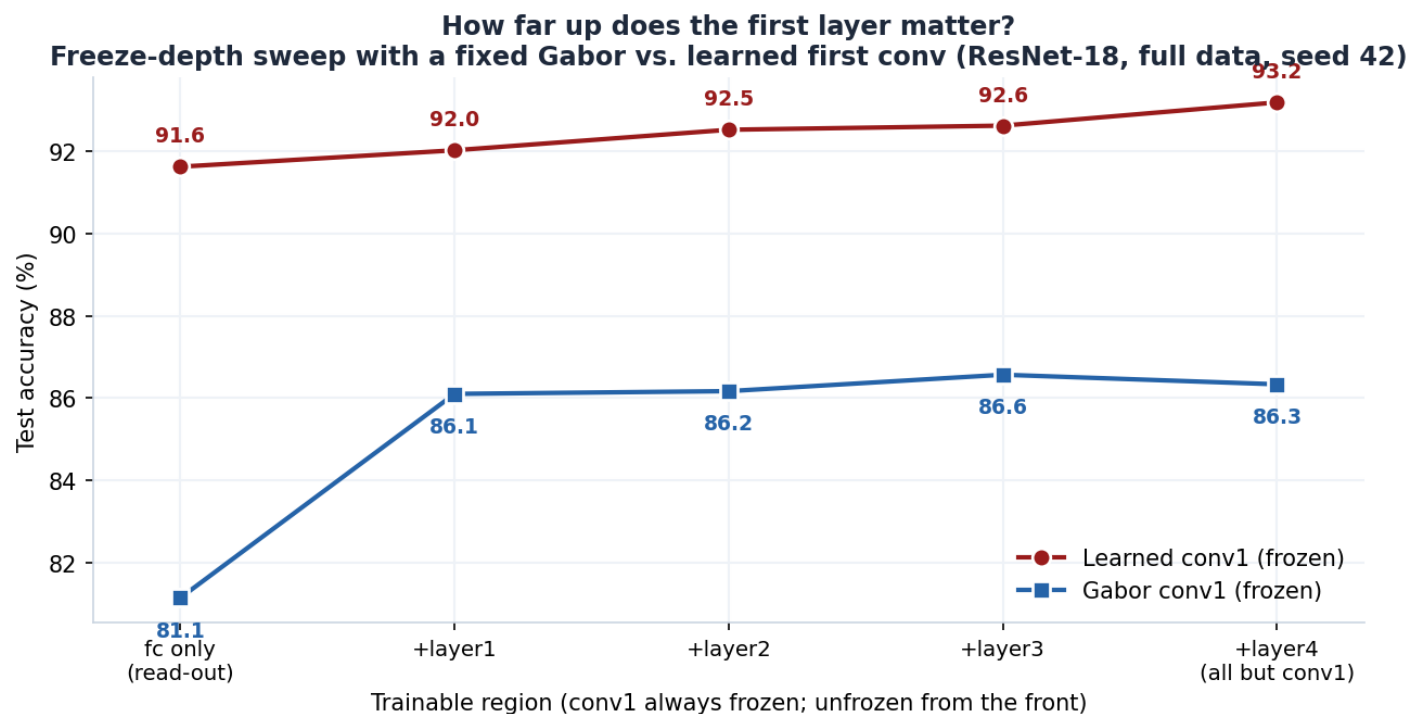
How much fine-tuning do you actually need?



- A bare linear probe already hits **91.3%** — frozen ImageNet features are nearly linearly separable.
- Unfreezing **layer4** adds ~1.6 pts: the boundary used in the main experiments.
- layer3, layer2, full fine-tune: **flat at 92.8%** — 3.6× the params, no gain.
- Most of pretraining's value needs **almost no fine-tuning**.

EXTENSION

How far up does the first layer reach?



- Swap **conv1** for a fixed **Gabor bank** (8 orientations $\times$ 4 freqs $\times$ 2 phases), freeze it, then unfreeze deeper stages one at a time.
- Gabor read-out: **81.1%** — about 10 pts below the learned first layer.
- Unfreezing just **layer1** recovers **+5 pts to 86.1%**; layer2–4 add almost nothing.
- First-layer influence reaches **~one stage up**. A ~7-pt gap to learned features never closes, even fully trainable.

The mistakes are the dataset's, not the method's

Pretrained-Frozen · test confusion (seed 100)

	B	F	G	M	Se	St
B	408	0	0	0	1	28
F	0	471	0	1	0	2
G	1	4	458	71	16	3
M	1	2	28	484	9	1
Se	2	0	2	3	503	0
St	26	0	0	2	0	473

- **Glacier ↔ Mountain** (71 errors): shared snow, ridges, rock texture — genuinely ambiguous.
- **Buildings ↔ Street** (26–28): streets contain buildings and vice-versa.
- The **same** confusions appear in all three conditions — just far larger for random init.
- Errors track **visual overlap in the data**, not a broken training setup.

What the control actually proved

01 Pretraining is the feature quality, not the parameter count

Capacity-matched, pretrained features add **+26.7 points** over random frozen features. The win is the representations themselves.

02 Capacity helps, but can't close the gap

Unfreezing everything (Random-Full) recovers to 84.7% — better than random-frozen, yet still ~9 points short of pretraining, at ~4× the compute.

03 For small data + tight budgets, transfer learning wins outright

Best accuracy and fastest convergence. The random-frozen control is what lets me say that with confidence.